Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 3: A Peek Beyond the Point

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	3 — A Peek Beyond the Point
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Longer-is-bigger ordering traps, misplaced/shifted decimal points when multiplying or dividing by powers of ten, interior-zero place shifts, and chained add/subtract/multiply slips are all baited with computable wrong answers. Do not write on this paper — mark answers on your answer sheet.

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1. A number has 3 in the tens place, 6 in the tenths place, and 8 in the thousandths place; every other place up to the thousandths is zero. The number is:

(A) 30.68 (B) 3.608
(C) 30.608 (D) 30.0608

2. Written as a decimal, 2 + 7/100 + 5/1000 is:

(A) 2.75 (B) 2.0705
(C) 2.075 (D) 2.705

3. Which of these four numbers is the LARGEST: 0.7, 0.65, 0.689, 0.6991?

(A) 0.6991 (B) 0.689
(C) 0.7 (D) 0.65

4. When 4.7 is multiplied by 100, the result is:

(A) 47 (B) 470
(C) 4700 (D) 0.047

5. Compute 6.5 - 2.78 + 1.4.

(A) 5.12 (B) 5.32
(C) 2.32 (D) 4.72

6. Three pens cost 12.50 rupees each and one notebook costs 27.75 rupees. The total bill is:

(A) 40.25 rupees (B) 52.75 rupees
(C) 65.25 rupees (D) 62.25 rupees

7. In 58.704, the place value of the digit 7 minus the place value of the digit 4 equals:

(A) 0.696 (B) 0.704
(C) 0.3 (D) 0.66

8. Which decimal lies exactly halfway between 3.4 and 3.5 on the number line?

(A) 3.9 (B) 3.45
(C) 3.405 (D) 3.5

9. A ribbon 0.6 m long is multiplied in length 4 times for a banner. The banner uses:

(A) 2.4 m (B) 0.24 m
(C) 24 m (D) 0.64 m

10. Written in simplest form, the decimal 0.375 equals:

(A) 375/100 (B) 3/80
(C) 375/1000 (D) 3/8

11. When 350 is divided by 1000, the result is:

(A) 3.5 (B) 0.035
(C) 35 (D) 0.35

12. A wire is 1.25 m long. Expressed in centimetres (1 m = 100 cm) it is:

(A) 1250 cm (B) 125 cm
(C) 12.5 cm (D) 0.0125 cm

13. How many thousandths are there altogether in 0.42?

(A) 42 (B) 420
(C) 4200 (D) 42000

14. Arrange from smallest to largest: 0.5, 0.45, 0.504, 0.054.

(A) 0.054, 0.45, 0.504, 0.5 (B) 0.45, 0.054, 0.5, 0.504
(C) 0.054, 0.45, 0.5, 0.504 (D) 0.054, 0.5, 0.45, 0.504

15. From a 5 m roll of tape, pieces of 1.45 m, 0.8 m and 1.25 m are cut. The length left is:

(A) 1.6 m (B) 2.5 m
(C) 1.5 m (D) 1.45 m

16. A toy costs 84.50 rupees. Buying 3 of them and paying with a 300-rupee amount, the change is:

(A) 46.50 rupees
(B) 215.50 rupees
(C) 46.05 rupees
(D) 43.50 rupees

17. A plank 3.6 m long is cut into 8 equal pieces. Each piece measures:

(A) 0.45 m
(B) 0.4 m
(C) 4.5 m
(D) 0.36 m

18. The sum 2.96 + 1.07 rounded to the nearest whole number is:

(A) 4
(B) 3
(C) 5
(D) 4.03

19. The fraction 9/4 written as a decimal is:

(A) 2.4
(B) 9.4
(C) 0.225
(D) 2.25

20. A bottle holds 1.5 litres. In millilitres (1 L = 1000 mL) this is:

(A) 150 mL
(B) 15000 mL
(C) 1500 mL
(D) 1.5 mL

21. On a number line marked in hundredths, which point is NEAREST to 6.5: 6.43, 6.56, 6.4 or 6.6?

(A) 6.43
(B) 6.6
(C) 6.4
(D) 6.56

22. Arrange from largest to smallest: 0.909, 0.99, 0.099, 0.9.

(A) 0.909, 0.99, 0.9, 0.099
(B) 0.99, 0.909, 0.9, 0.099
(C) 0.99, 0.909, 0.099, 0.9
(D) 0.099, 0.9, 0.909, 0.99

23. When 0.06 is multiplied by 1000, the answer is:

(A) 6
(B) 60
(C) 600
(D) 0.6

24. Compute 100 - 37.45 - 8.9.

(A) 54.55
(B) 53.65
(C) 62.55
(D) 53.55

25. Rounded to the nearest tenth, the difference 5.2 - 0.36 is:

(A) 4.9
(B) 5.0
(C) 4.84
(D) 4.8

26. A number x has the property that x divided by 10 equals 0.07. The number x is:

(A) 0.007
(B) 7
(C) 0.7
(D) 0.0007

27. A pencil costs 7.25 rupees and a pen costs 11.50 rupees. Buying 2 pencils and 1 pen and paying with a 50-rupee note, the change is:

(A) 31.25 rupees
(B) 23.75 rupees
(C) 24.00 rupees
(D) 25.00 rupees

28. A bag of rice weighs 2.5 kg. The total weight of 6 such bags is:

(A) 1.5 kg
(B) 150 kg
(C) 12.5 kg
(D) 15 kg

29. In kilometres, 2350 metres (1 km = 1000 m) is:

(A) 23.5 km
(B) 235 km
(C) 0.235 km
(D) 2.35 km

30. When 8.4 is divided by 100, the result is:

(A) 0.84
(B) 0.0084
(C) 0.084
(D) 84

31. The decimal 'two and four hundred six thousandths' written in figures is:

(A) 2.460
(B) 2.0406
(C) 2.46
(D) 2.406

32. A tank holds 4 L. After 1.65 L and 0.85 L are poured out and then 0.5 L is added back, the tank holds:

(A) 1 L
(B) 2.5 L
(C) 1.5 L
(D) 2 L

33. Which decimal lies exactly halfway between 2.07 and 2.09?

(A) 2.08
(B) 2.8
(C) 2.075
(D) 2.085

34. Place in increasing order: 1.1, 1.01, 1.101, 1.011.

(A) 1.01, 1.011, 1.1, 1.101
(B) 1.01, 1.011, 1.101, 1.1
(C) 1.011, 1.01, 1.1, 1.101
(D) 1.01, 1.1, 1.011, 1.101

35. Rounded to the nearest whole number, the product 1.9 x 3 is:

(A) 5
(B) 6
(C) 3
(D) 5.7

36. A jug had some juice. After 0.45 L was added it held 1.2 L. The amount of juice originally in the jug was:

(A) 1.65 L
(B) 0.75 L
(C) 0.85 L
(D) 0.65 L

37. Cloth costs 124.50 rupees per metre. The cost of 3 metres is:

(A) 124.50 rupees
(B) 372.50 rupees
(C) 373.50 rupees
(D) 363.50 rupees

38. How many tenths altogether are there in 3.6?

(A) 6
(B) 36
(C) 360
(D) 3.6

39. The decimal 0.0125 multiplied by 100 gives:

(A) 12.5
(B) 0.125
(C) 125
(D) 1.25

40. Which fraction equals the decimal 0.16 in simplest form?

(A) 4/25
(B) 16/100
(C) 8/50
(D) 1/16

41. A relay team's three laps are 12.4 s, 11.85 s and 12.05 s. The total time is:

(A) 35.3 s
(B) 36.4 s
(C) 36.13 s
(D) 36.3 s

42. When a decimal is multiplied by 1000 the point moves three places right; when divided by 100 it moves two places left. Starting from 5.2, doing both in turn gives:

(A) 52
(B) 520
(C) 5.2
(D) 0.52

43. The number 7.5 lies between which pair when each is multiplied by 10?

(A) between 7.4 and 7.6
(B) between 74 and 76 (since 7.5 x 10 = 75)
(C) between 740 and 760
(D) between 0.74 and 0.76

44. Rounded to the nearest tenth, the quotient 3.7 / 2 is:

(A) 1.8
(B) 2.0
(C) 1.85
(D) 1.9

45. Counting back by hundredths from 3.02, the third number reached is:

(A) 3.00
(B) 2.92
(C) 2.72
(D) 2.99

46. A digit 5 has place value 0.05 in one number and place value 5 in another. The first place value is what fraction of the second?

(A) 1/10
(B) 1/1000
(C) 1/100
(D) 1/5

47. Compute 9 - 4.56 - 2.7.

(A) 1.84
(B) 2.26
(C) 11.74
(D) 1.74

48. A length of 0.875 m is needed for each shelf. The total length for 8 shelves is:

(A) 7.875 m
(B) 70 m
(C) 7 m
(D) 0.875 m

49. Which fraction does NOT give a terminating (ending) decimal?

(A) 5/12 (B) 7/16
(C) 9/40 (D) 3/25

50. By how much is 1.005 greater than 0.95?

(A) 0.045 (B) 0.055
(C) 0.95 (D) 0.155

51. A scale shows 2.4 kg, but the true mass is 7 hundredths of a kg more. The true mass is:

(A) 2.7 kg (B) 3.1 kg
(C) 2.47 kg (D) 2.407 kg

52. Which trial is the SHORTEST: 200.6 m, 200.06 m, 200.5 m, 200.65 m?

(A) 200.5 m (B) 200.06 m
(C) 200.65 m (D) 200.6 m

53. Rounded to the nearest whole number, the sum 150.5 + 0.49 becomes:

(A) 150 (B) 151
(C) 152 (D) 150.99

54. A digit moved from the hundredths place to the tenths place has its value multiplied by:

(A) 10 (B) 100
(C) 1/10 (D) 1000

55. The greatest decimal that can be formed using each of the digits 6, 2 and 0 once, with exactly one digit before the decimal point, is:

(A) 6.20 (B) 6.02
(C) 2.60 (D) 0.62

56. Compute 12.5 x 4 - 18.6.

(A) 68.6 (B) 49.4
(C) 31.4 (D) 13.4

57. How many hundredths altogether are there in 2.05?

(A) 205 (B) 25
(C) 20.5 (D) 2050

58. Three identical books are stacked to a height of 9.6 cm. After adding one more identical book, the stack height is:

(A) 10.6 cm (B) 12.8 cm
(C) 9.6 cm (D) 38.4 cm

59. Sara had 100 rupees. She bought 3 items at 12.75 rupees each. The money left is:

(A) 61.75 rupees (B) 87.25 rupees
(C) 61.25 rupees (D) 62.25 rupees

60. The total of 0.625, 0.375 and 0.25 is:

(A) 1.0

(B) 0.125

(C) 1.5

(D) 1.25

Solutions — Mathematics (Class 7), Chapter 3: A Peek Beyond the Point — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	D	31	D	41	D	51	C
2	C	12	B	22	B	32	D	42	A	52	B
3	C	13	B	23	B	33	A	43	B	53	B
4	B	14	C	24	B	34	A	44	D	54	A
5	A	15	C	25	D	35	B	45	D	55	A
6	C	16	A	26	C	36	B	46	C	56	C
7	A	17	A	27	C	37	C	47	D	57	A
8	B	18	A	28	D	38	B	48	C	58	B
9	A	19	D	29	D	39	D	49	A	59	A
10	D	20	C	30	C	40	A	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Tens = 30, tenths = 0.6, hundredths = 0 (stated), thousandths = 0.008. Sum = 30 + 0.6 + 0.008 = 30.608. Writing 30.68 wrongly drops the empty hundredths place, pushing the 8 from thousandths up into hundredths.

2. (C) 7/100 = 0.07 and 5/1000 = 0.005, so 2 + 0.07 + 0.005 = 2.075. Reading 7/100 as 7 tenths gives the wrong 2.75.

3. (C) Compare from the tenths: 0.7 has 7 tenths while the other three all have 6 tenths, so 0.7 is largest. Picking 0.6991 falls for the longer-is-bigger trap of counting digits instead of place value.

4. (B) Multiplying by 100 shifts the decimal point two places right: 4.7 -> 47. -> 470. Shifting only one place (treating x100 as x10) gives the wrong 47.

5. (A) 6.5 - 2.78 = 3.72, then 3.72 + 1.4 = 5.12. Adding before subtracting wrongly (6.5 + 1.4 - 2.78 done as 6.5 - (2.78+1.4)) or right-aligning digits produces 5.32 or 2.32.

6. (C) Three pens = 3 x 12.50 = 37.50, plus 27.75 = 65.25. Adding only one pen (12.50 + 27.75 = 40.25) ignores that there are three pens.

7. (A) Place value of 7 is 0.7 (tenths); place value of 4 is 0.004 (thousandths). 0.7 - 0.004 = 0.696. Subtracting the bare digits 7 - 4 = 3 and reading it as 0.3 ignores the places.

8. (B) The gap is 0.1; half of 0.1 is 0.05, so the midpoint is 3.4 + 0.05 = 3.45. Adding 3.4 and 3.5 to get 6.9 and halving carelessly, or thinking half of 0.1 is 0.005, gives the wrong answers.

9. (A) 0.6 x 4 = 2.4. Multiplying 6 x 4 = 24 then misplacing the point as 0.24 keeps the point too far left.

10. (D) 0.375 = 375/1000; dividing top and bottom by 125 gives 3/8. Leaving it as 375/1000 is correct but not simplest; 3/80 comes from cancelling only one zero.

11. (D) Dividing by 1000 shifts the decimal point three places left: 350. -> 35.0 -> 3.50 -> 0.350 = 0.35. Shifting only two places gives the wrong 3.5.

12. (B) 1.25 m x 100 = 125 cm (point moves two places right). Moving the point three places, as if 1 m = 1000 cm, gives the wrong 1250 cm.

13. (B) 0.42 = 0.420 = 420/1000, so there are 420 thousandths. Counting only the hundredths (42) forgets that each hundredth is 10 thousandths.

14. (C) Compare tenths: 0.054 has 0 tenths (smallest); then 0.45 (4 tenths); then 0.5 and 0.504 both have 5 tenths, and 0.5 = 0.500 < 0.504. So 0.054, 0.45, 0.5, 0.504. Placing 0.504 below 0.5 falls for longer-is-bigger reversed.

15. (C) Cut total = 1.45 + 0.8 + 1.25 = 3.5 m; left = 5 - 3.5 = 1.5 m. Mis-adding 0.8 as 0.08 gives a cut of 2.78 and a wrong remainder.

16. (A) Three toys = 3 x 84.50 = 253.50; change = 300 - 253.50 = 46.50. Subtracting only one toy (300 - 84.50 = 215.50) ignores the quantity 3.

17. (A) 3.6 / 8 = 0.45 (since 36 / 8 = 4.5, then 3.6 / 8 = 0.45). Dividing 3.6 by 8 and dropping a place gives the wrong 0.4 or shifting the point gives 4.5.

18. (A) 2.96 + 1.07 = 4.03; rounded to the nearest whole number that is 4. Rounding each number first (3 + 1 = 4 is fine, but 3.0 + 1.0 then truncating) or chopping to 3 mishandles the carry.

19. (D) 9/4 = 9 divided by 4 = 2.25 (since 4 x 2 = 8, remainder 1 -> 1.00/4 = 0.25). Reading 9/4 as 9.4 or treating it as 9 tenths over 4 gives the wrong answers.

20. (C) 1.5 L x 1000 = 1500 mL (point moves three places right). Moving the point only two places, as if 1 L = 100 mL, gives the wrong 150 mL.

21. (D) Distances from 6.5: |6.43-6.5| = 0.07, |6.56-6.5| = 0.06, |6.4-6.5| = 0.1, |6.6-6.5| = 0.1. Smallest is 0.06, so 6.56. Judging by digit count instead of distance favours 6.43.

22. (B) All but 0.099 have 9 tenths. Among those, compare hundredths: 0.99 has 9, 0.909 has 0, 0.9 has 0; 0.909 vs 0.9 -> 0.909 > 0.900. So 0.99 > 0.909 > 0.9 > 0.099. Counting digits puts 0.909 first wrongly.

23. (B) Multiplying by 1000 shifts the point three places right: 0.06 -> 0.6 -> 6. -> 60. Shifting only two places gives the wrong 6.

24. (B) 100 - 37.45 = 62.55, then 62.55 - 8.9 = 53.65. Subtracting 8.9 as 8.09 gives 54.46-type errors; forgetting the second subtraction leaves 62.55.

25. (D) 5.2 - 0.36 = 4.84; the hundredths digit 4 is below 5, so it rounds down to 4.8. Rounding 4.84 up to 4.9 mishandles the 'round down' rule.

26. (C) If x / 10 = 0.07, then x = 0.07 x 10 = 0.7 (point moves one place right). Dividing again instead of multiplying gives the wrong 0.007.

27. (C) Cost = 2 x 7.25 + 11.50 = 14.50 + 11.50 = 26.00; change = 50 - 26 = 24.00. Buying only one pencil (7.25 + 11.50 = 18.75; change 31.25) ignores the second pencil.

28. (D) 2.5 x 6 = 15. Computing 25 x 6 = 150 then misplacing the point as 1.5 puts the decimal too far left.

29. (D) 2350 / 1000 = 2.35 (point moves three places left). Moving only two places gives the wrong 23.5 km.

30. (C) Dividing by 100 shifts the point two places left: 8.4 -> 0.84 -> 0.084. Shifting only one place gives the wrong 0.84.

31. (D) Four hundred six thousandths = 406/1000 = 0.406, so the number is 2.406. Writing 2.460 reverses the hundredths and thousandths; 2.46 drops the thousandths place entirely.

32. (D) Poured out = 1.65 + 0.85 = 2.5; remaining = 4 - 2.5 = 1.5; add back 0.5 -> 2.0 L. Forgetting to add the 0.5 back leaves 1.5 L.

33. (A) Gap is 0.02; half is 0.01, so midpoint = 2.07 + 0.01 = 2.08. Thinking the midpoint of 7 and 9 hundredths needs another digit produces 2.075 or 2.085.

34. (A) Tenths: 1.01 and 1.011 have 0 tenths; 1.1 and 1.101 have 1 tenth. Among the first pair 1.010 < 1.011; among the second 1.100 < 1.101. So 1.01, 1.011, 1.1, 1.101. Ranking by digit count puts 1.101 before 1.1 wrongly.

35. (B) 1.9 x 3 = 5.7; the tenths digit 7 is 5 or more, so it rounds up to 6. Rounding 1.9 down to 1 first (1 x 3 = 3) or chopping 5.7 to 5 ignores the carry.

36. (B) Original + 0.45 = 1.2, so original = 1.2 - 0.45 = 0.75 L. Adding instead of subtracting gives the wrong 1.65 L.

37. (C) 3 x 124.50 = 373.50. Adding three half-rupees should give 1.50 over 372 (3 x 124 = 372, plus 3 x 0.50 = 1.50 -> 373.50); dropping a carry yields 372.50.

38. (B) 3 wholes = 30 tenths and 0.6 = 6 tenths, so 36 tenths in all. Counting only the digit after the point (6) ignores the 30 tenths inside the 3 wholes.

39. (D) Multiplying by 100 moves the point two places right: 0.0125 -> 0.125 -> 1.25. Moving three places (as if x1000) gives the wrong 12.5.

40. (A) 0.16 = 16/100; dividing by 4 gives 4/25. Leaving it as 16/100 or only halving to 8/50 is not simplest form.

41. (D) 12.4 + 11.85 + 12.05 = 36.30 = 36.3 s. Right-aligning the digits instead of the points (treating 12.4 as 12.04) gives 36.13.

42. (A) 5.2 x 1000 = 5200; 5200 / 100 = 52 (net three minus two = one place right). Net shift gives one place right of 5.2 = 52. Treating the net as two places gives 520.

43. (B) 7.5 x 10 = 75, which lies between 74 and 76. Forgetting to apply the x10 shift keeps the comparison around 7.5, giving 'between 7.4 and 7.6'.

44. (D) 3.7 / 2 = 1.85; the hundredths digit 5 rounds the tenths up: 1.85 -> 1.9. Rounding down to 1.8 ignores the round-half-up rule.

45. (D) 3.02 -> 3.01 (1st) -> 3.00 (2nd) -> 2.99 (3rd). Stopping at the second step gives 3.00; subtracting tenths instead of hundredths gives 2.72-type errors.

46. (C) 0.05 / 5 = 0.01 = 1/100. Going only one place (tenths to ones) gives the wrong 1/10.

47. (D) 9 - 4.56 = 4.44, then 4.44 - 2.7 = 1.74. Subtracting 2.7 as 2.07 gives 2.37; mis-aligning leaves 1.84.

48. (C) 0.875 x 8 = 7.000 = 7 m (since 875 x 8 = 7000). Adding 7 to 0.875 wrongly gives 7.875.

49. (A) A fraction terminates only if its denominator (in lowest terms) has just 2s and 5s as prime factors. 12 = 4 x 3 has a factor 3, so 5/12 does not terminate. 16, 40 (= 8 x 5) and 25 have only 2s and 5s, so those terminate.

50. (B) 1.005 - 0.95 = 1.005 - 0.950 = 0.055. Subtracting right-aligned (treating 0.95 as 0.095) gives the wrong 0.045-type result.

51. (C) 7 hundredths = 0.07, so 2.4 + 0.07 = 2.47 kg. Adding 7 tenths (0.7) gives the wrong 3.1; placing the 7 in thousandths gives 2.407.

52. (B) All share 200.; compare tenths: 200.06 has 0 tenths, the smallest, so it is shortest. Judging by total digit count would wrongly flag 200.65.

53. (B) 150.5 + 0.49 = 150.99; the tenths digit 9 is 5 or more, so it rounds up to 151. Rounding each addend first (150.5 -> 151 wrongly via early rounding, or chopping 150.99 to 150) misses that the true sum 150.99 is just below 151 and still rounds up to 151.

54. (A) Each step left multiplies the place value by 10 (hundredths 0.01 -> tenths 0.1 is x10). Confusing the place names with hundreds/tens suggests x100.

55. (A) To be greatest, put the biggest digit (6) before the point, then the next biggest (2) in the tenths place: 6.20. Putting 0 in the tenths gives the smaller 6.02.

56. (C) 12.5 x 4 = 50, then 50 - 18.6 = 31.4. Multiplying after subtracting (12.5 x (4 - 18.6)) or mis-subtracting gives 49.4 or 13.4.

57. (A) 2.05 = 205/100, so there are 205 hundredths. Counting 2 wholes as only 20 hundredths and adding the 5 gives 25, forgetting that 1 whole is 100 hundredths.

58. (B) One book = 9.6 / 3 = 3.2 cm; four books = 9.6 + 3.2 = 12.8 cm. Adding 1 cm instead of one book's height gives the wrong 10.6 cm.

59. (A) Cost = 3 x 12.75 = 38.25; left = 100 - 38.25 = 61.75. Buying only one item (100 - 12.75 = 87.25) ignores the quantity 3.

60. (D) 0.625 + 0.375 = 1.000, then + 0.25 = 1.25. Stopping at the first pair gives 1.0; right-aligning the 0.25 gives errors near 0.125.